NEXVIA AEGIS V9-M4 — Owner Decision Record

Phase 22: Formal Pre-Execution Owner Decision Gate

Review of Report 21 Consistency Audit and Production Holdout Decision

Autonomous Verification Subsystem (Google DeepMind Antigravity)

Git Branch: v9.0-runtime-assurance · Base Commit: c271d76 · Date: October 26, 2026

CRITICAL NON-AUTHORIZATION INVARIANT:
This document records the formal owner decision regarding the statistical protocol corrections in Report 21 and the evaluation targets for Milestone V9-M4. **This document does NOT constitute authorization to execute Milestone V9-M4, nor does it provide or invoke any execution token.** Live execution of Milestone V9-M4 remains strictly **NOT AUTHORIZED**. All production holdout paths remain uninstantiated and locked.

1 Formal Decision Summary

Decision Field	Formal Determination	Operational Scope
1. Acceptance of Report 21 Corrections	**ACCEPT**	Accepts all 5 source-of-truth corrections in Report 21, formally superseding narrative transcription errors in Report 20.
2. Confirmation of Registered Targets	**CONFIRMED**	Confirms governing H_1–H_5 thresholds, sampling allocations, baseline identities, and H_3 S_{10} scope.
3. Production Holdout Decision	**NOT AUTHORIZED**	Production confirmatory holdout execution remains strictly locked and unauthorized.

Table 1: Formal Owner Decision Determinations for Milestone V9-M4.

2 Executive Summary

2.1 Executive Summary (English)

1. **Acceptance of Corrections (ACCEPT):** The project owner accepts the five source-of-truth reporting corrections documented in Report 21. Production code and preregistered protocol were verified to be in 100% agreement; discrepancies in Report 20 were purely narrative transcription errors.
2. **Confirmation of Targets (CONFIRMED):** Confirms governing thresholds: H_1 ($U_{\mathrm{CP},1} \leq 0.05$ on $M \geq 59$), H_4 ($\hat{C} \geq 0.35$), Scenario Holdout total ($N = 1,440$ episodes across 12 scenarios $\times$ 120 seeds), H_5 across all 12 scenarios ($N_s = 120$ each), and H_3 ($N = 1,200$ matched pairs strictly on S_{10}).
3. **Holdout Execution Status (NOT AUTHORIZED):** Confirmatory holdout evaluation remains safely locked. Acceptance of documentation corrections does not constitute execution authorization.
4. **Holdout Independence:** Confirmed that zero real holdout data was generated, loaded, or inspected. Reserved seed blocks (73000..74439 and 75000..75119) remain 100% untouched.

2.2 บทสรุปสำหรับผู้บริหาร (Executive Summary in Thai)

1. **การยอมรับข้อแก้ไขใน Report 21 (ACCEPT):** ยอมรับข้อแก้ไขเชิงรายงานทั้ง 5 ข้อใน Report 21 อย่างเป็นทางการ รหัสต้นฉบับการประเมินผลและพิธีสารมีความสอดคล้องกันอย่างสมบูรณ์แบบ โดยข้อผิดพลาดใน Report 20 เป็นเพียงความคลา
2. **การยืนยันเป้าหมายและเกณฑ์ทางสถิติ (CONFIRMED):** ยืนยันเกณฑ์การประเมินตามพิธีสารหลัก: H_1 (≤ 0.05 บน $M \geq 59$), H_4 (≥ 0.35), Scenario Holdout รวม 1,440 ตอน (12 ฉากทัศน์ $\times$ 120 ตอน), และ H_3 ประเมินเฉพาะกลุ่ม S_{10} จำนวน 1,200 คู่เทียบ
3. **สถานะการอนุมัติการรันจริง (NOT AUTHORIZED):** การประเมินผลชุดข้อมูลจริงใน Milestone V9-M4 ยังคงไม่อนุมัติให้ดำเนิน การยืนยันความถูกต้องของระเบียบวิธีไม่ถือเป็นการอนุญาตให้รันจริง

4. **ความบริสุทธิ์ของชุดข้อมูล Holdout:** ยืนยัน 100% ว่าไม่มีการสร้าง เข้าถึง หรือจำลองข้อมูลในชุด Holdout บล็อก seed สำรองยังคงไม่ถูกแตะต้อง

3 Governing Statistical Protocol and Decision Rules

Hypothesis	Target Partition	Estimand / Metric	Sample Size	Governing Pass Threshold
H_1 (Risk)	$\mathcal{D}_{\text{test}}^{\text{mix}}$	Selective Risk $R_{\text{sel}} = K/M$	$N = 1,440$	Exact 1-sided 95% Clopper–Pearson $U_{\text{CP},1} \leq \mathbf{0.05}$ on $M \geq 59$
H_2 (Incidence)	$\mathcal{D}_{\text{test}}^{\text{mix}}$	FS Incidence Reduction vs Baselines	1,440 pairs	Paired McNemar $p < \mathbf{0.05}$ with $n_{10} < n_{01}$ (vs V8.1 & V7)
H_3 (Utility)	$\mathcal{D}_{\text{test}}^{\text{scen}} \cap \mathcal{S}_{10}$	Paired TSR Difference vs Baselines	1,200 pairs	Student's t Wald $\underline{\text{LCB}}_{95} > -\mathbf{0.05}$ on S_{10} AND $\widehat{\text{TSR}}_{\text{nom}} \geq \mathbf{0.94}$
H_4 (Coverage)	$\mathcal{D}_{\text{test}}^{\text{mix}}$	Task Coverage Floor $\hat{C} = M/N$	$N = 1,440$	Sample proportion point estimate $\hat{C} \geq \mathbf{0.35}$ ($M \geq 504$)
H_5 (Safety)	$\mathcal{D}_{\text{test}}^{\text{scen}}$	Worst-Case Scenario Safety	1,440 eps	Berger IUT $\max_{s \in \mathcal{S}_{12}}(p_{\text{inc},s}) \leq \mathbf{0.05}$ ($N_s = 120$)

Table 2: Governing Confirmatory Decision Rules Confirmed for Milestone V9-M4.

4 Owner Rationale and Outstanding Conditions

- **Rationale for Accepting Report 21 Corrections:** The protocol consistency audit proved 100% agreement between the implementation (`experiments/v9_evaluate_holdout.py`) and the preregistered protocol (`docs/v9_experiment_protocol.md`). Regression suites pass 62/62 (preflight) and 173/173 (full repository). The five discrepancies in Report 20 were clerical transcription errors.
- **Rationale for NOT AUTHORIZED Status:** Protocol verification is a necessary quality prerequisite, but does not substitute for an independent owner authorization action outside the audit documentation workflow.
- **Outstanding Conditions for Future Authorization:**
  1. Owner executive review and sign-off on this Decision Record.
  2. Explicit manual injection of the secret execution token into the runtime execution command.
  3. Dedicated single-shot execution environment preventing collision with existing output directories.

5 Repository Identity and Holdout Verification

- **Git Branch:** `v9.0-runtime-assurance` · **Commit:** `c271d76`.
- **Certified Frozen Policy:** `results/v9/cal_r2/selected_policy.json` (`V9_CAND_1_TIER1_STRICTEST`).
- **Raw SHA-256:** `5a89d6061b379f1b34ab47be0deef8d63180d3e4c21d18c478bc7e451f5c84e2`.
- **Canonical SHA-256:** `90dc5c0068bb62187415bba57fbf9cafc1f18e111ad5790b0214961490902034`.
- **Holdout Isolation:** Confirmed 100% absent. Reserved seed blocks (73000..74439 and 75000..75119) untouched.